Academic year 2024-2025 (Even Sem)

DEPARTMENTS OF ISE & CSE

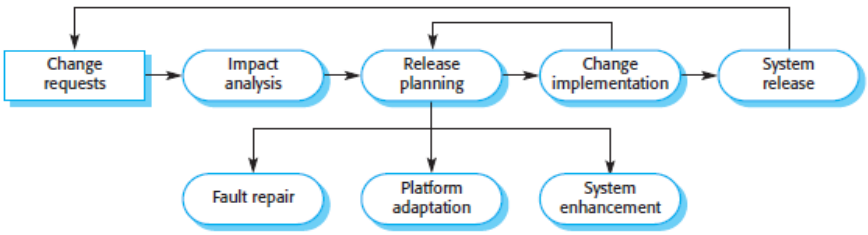
Date: /05/2025	**CIE - II**	**Max. Marks:60**
Semester: VI	**UG**	**Duration: 2 Hour**
Course Title: Software Engineering with Agile Technologies		**Course Code: IS364TA**

Sl. No.	Questions	M	BT	CO
	PART A(QUIZ)			
1.1	_____, _____ and _____ are the distinct types of testing.	1	1	1
1.2	Open-source licenses are variants of one of three general models they are _____ , _____ and _____.	1	1	2
1.3	Differentiate between the software reengineering and refactoring.	2	2	3
1.4	Write a block diagram activity depicting general model of the software evolution process.	2	3	3
1.5	List the six stages in the acceptance testing process.	2	3	1
1.6	List different levels of Software reuse.	2	2	2

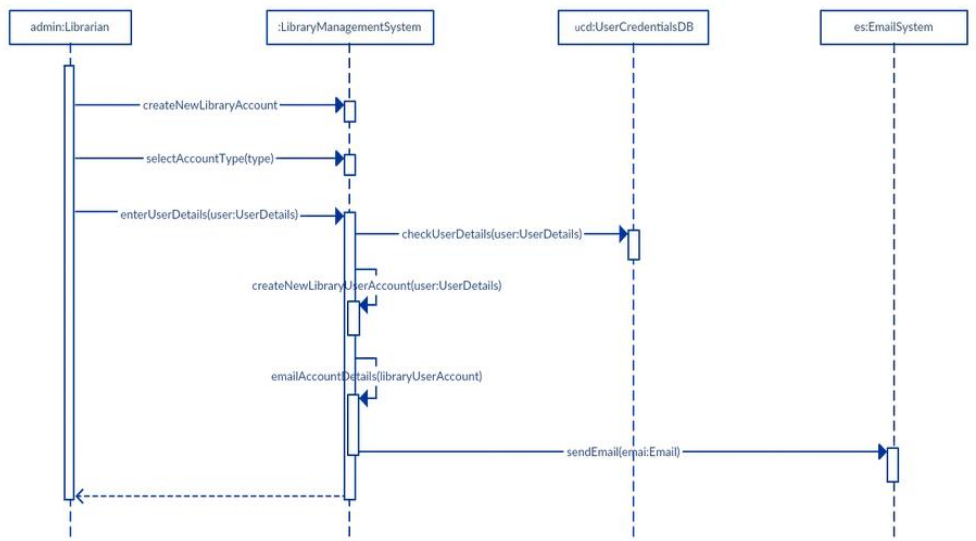
PART B(TEST)				
1a.	List and explain any 3 commonly used Architectural patterns.	6	2	1
1b.	Draw diagrams showing a conceptual view and a process view of the architectures of the following systems: a. A computer-controlled video conferencing system that allows video, audio, and computer data to be visible to several participants at the same time. b. A robot floor-cleaner that is intended to clean relatively clear spaces such as corridors. The cleaner must be able to sense walls and other obstructions.	4	1	1
2a.	You have been asked to test a method called catWhiteSpace in a "Paragraph" object that, within the paragraph, replaces sequences of blank characters with a single blank character. Identify testing partitions for this example and derive a set of tests for the catWhiteSpace method.	6	3	3
2b.	List and explain four benefits of test-driven development.	4	2	3
3a.	Create New Online Library Account, the objects would be Librarian, Online Library Management system, User credentials database, Email system. The detailed description of the interactions that would occur between the objects are given below. • The librarian requests the system to create a new online library account • The librarian then selects the library user account type • The librarian enters the user's details • The user's details are checked using the user Credentials Database • The new library user account is created • A summary of the new account's details is then emailed to the user With the above information draw the sequence diagram and explain.	6	4	4
3b.	Differentiate between three different types of software maintenance.	4	3	2
4a.	Analyse two main reasons why the most companies are reluctance to spend more on software development to reduce longer-term maintenance costs.	5	2	3
4b.	Some people argue that developers should not be involved in testing their own code but that all testing should be the responsibility of a separate team. Give arguments for and against testing by the developers themselves.	5	2	2
5a.	With neat diagram explain logical parts of a legacy system and their relationships.	6	3	2
5b.	Write the State diagram for the Weather station and explain.	4	4	4

Academic year 2024-2025 (Even Sem)

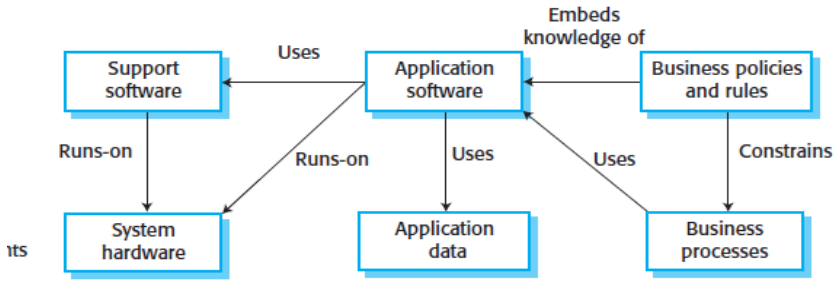
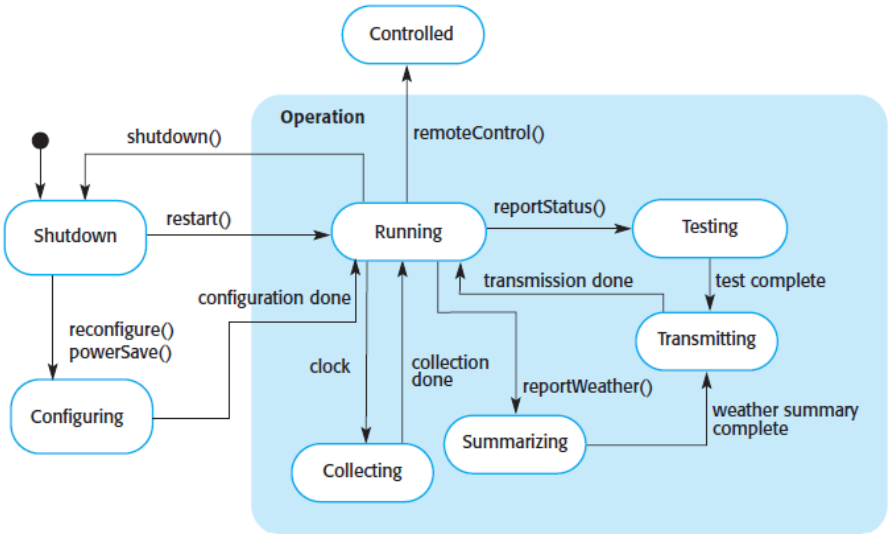
Date: 29/04/2025	**CIE – 1 SCHEME AND SOLUTION**	**Max. Marks:60**
Semester: VI	**UG**	**Duration: 2 Hour**
Course Title: Software Engineering with Agile Technologies		**Course Code: IS364TA**

Sl. No.	Questions	M
1.1	component testing, system testing, and release testing;	1
1.2	The GNU General Public License (GPL). The GNU Lesser General Public License (LGPL), The Berkley Standard Distribution (BSD) License.	1
1.3	Differentiate between the software reengineering and refactoring. • Software reengineering is concerned with restructuring and redocumenting software to make it easier to understand and change. • Refactoring, making small program changes that preserve functionality, can be thought of as preventative maintenance.	2
1.4		2
1.5	Define acceptance criteria, Plan acceptance testing, Derive acceptance, Run acceptance tests, Negotiate test results, Reject/accept system.	2
1.6	*abstraction level, The object level, The component level, The system level*	2

1a.	List and explain any 3 commonly used Architectural patterns. model-view-controller, layered architecture, repository, client–server, and pipe and filter. ---- list + explanation of any 3 patterns -----2+4 marks	6
1b.	Draw diagrams showing a conceptual view and a process view of the architectures of the following systems:---------- 2+ 2 marks	4
2a.	To test the** 'catwhitespace'** method in a 'paragraph' object, we need to identify testing partitions and create a set of tests. The method replaces sequences of ___ (whitespace) characters with a single blank character. Based on the information provided, here are some testing partitions and a set of tests for the 'catwhitespace' method 1. Paragraph with no whitespace. 2. Paragraph with single spaces between words, no modification needed. 3. Paragraph with multiple contiguous spaces between some words. 4. Paragraph with tab characters. 5. Paragraph with mixed whitespace characters (tabs, spaces, newlines). 6. Paragraph with whitespace at the beginning and/or end. For each of these partitions, one would derive a set of tests: • Feed a paragraph with no whitespaces and check if the output remains unchanged. • Provide a correctly spaced paragraph and verify no changes occur after method execution. • Insert multiple spaces between words and ensure they collapse into a single space. • Include tabs in the paragraph and test if they are converted to single spaces. • Use a combination of whitespace types and check for their proper conversion to single spaces. • Test paragraphs with leading and trailing whitespace to confirm they are effectively trimmed.	6

2b.	List and explain four benefits of test-driven development are: 1. *Code coverage* In principle, every code segment that you write should have at least one associated test. Therefore, you can be confident that all of the code in the system has actually been executed. Code is tested as it is written, so defects are discovered early in the development process. 2. *Regression testing* A test suite is developed incrementally as a program is developed. You can always run regression tests to check that changes to the program have not introduced new bugs. 3. *Simplified debugging* When a test fails, it should be obvious where the problem lies. The newly written code needs to be checked and modified. You do not need to use debugging tools to locate the problem. Reports of the use of TDD suggest that it is hardly ever necessary to use an automated debugger in test-driven development (Martin 2007). 4. *System documentation* The tests themselves act as a form of documentation that describe what the code should be doing. Reading the tests can make it easier to understand the code.	4
3a.	With the above information draw the sequence diagram and explain.  	6
3b.	There are three different types of software maintenance: 1. *Fault repairs to fix bugs and vulnerabilities.* Coding errors are usually relatively cheap to correct; design errors are more expensive because they may involve rewriting several program components. Requirements errors are the most expensive to repair because extensive system redesign may be necessary. 2. *Environmental adaptation to adapt the software to new platforms and environments.* This type of maintenance is required when some aspect of a system's environment, such as the hardware, the platform operating system, or other support software, changes. Application systems may have to be modified to cope with these environmental changes.	4

	3. *Functionality addition to add new features and to support new requirements.* This type of maintenance is necessary when system requirements change in response to organizational or business change. The scale of the changes required to the software is often much greater than for the other types of maintenance.	
4a.	Analyse two main reasons why the most companies are reluctance to spend more on software development to reduce longer-term maintenance costs. There are 1. Companies set out quarterly or annual spending plans, and managers are incentivized to reduce short-term costs. Investing in maintainability leads to short term cost increases, which are measurable. However, the long-term gains can't be measured at the same time, so companies are reluctant to spend money on something with an unknown future return. 2. Developers are not usually responsible for maintaining the system they have developed. Consequently, they don't see the point of doing additional work that might reduce maintenance costs, as they will not get any benefit from it.	5
4b.	Some people argue that developers should not be involved in testing their own code but that all testing should be the responsibility of a separate team. Give arguments for and against testing by the developers themselves. ----arguments for 2.5 + against 2.5 marks	5
5a.	With neat diagram explain logical parts of a legacy system and their relationships. *System hardware* Legacy systems may have been written for hardware that is no longer available, that is expensive to maintain, and that may not be compatible with current organizational IT purchasing policies. 2. *Support software* The legacy system may rely on a range of support software from the operating system and utilities provided by the hardware manufacturer through to the compilers used for system development. Again, these may be obsolete and no longer supported by their original providers. 3. *Application software* The application system that provides the business services is usually made up of a number of application programs that have been developed at different times. Some of these programs will also be part of other application software systems. 4. *Application data* These data are processed by the application system. In many legacy systems, an immense volume of data has accumulated over the lifetime of the system. This data may be inconsistent, may be duplicated in several files, and may be spread over a number of different databases. 5. *Business processes* These processes are used in the business to achieve some business objective. An example of a business process in an insurance company would be issuing an insurance policy; in a manufacturing company, a business process would be accepting an order for products and setting up the associated manufacturing process. Business processes may be designed around a legacy system and constrained by the functionality that it provides. *Business policies and rules* These are definitions of how the business.	5

		
5b.	Write the State diagram for the Weather station and explain. 	5